

Pre-Class Assignment: Chapter 5

Complements to Prediction

ECON 470 — Microeconomics of Artificial Intelligence

Spring 2026

Reading: Gans, Chapter 5 (Complements to Prediction), introduction and Sections 5.1–5.2

Instructions: Answer each question in 1–3 sentences. These questions are designed to be straightforward if you did the reading.

You must submit this assignment in person. The assignment must be handwritten. No photocopies will be accepted.

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1. In Gans’s Chapter 5 model, what is **judgment**? How does it differ from **prediction**?
 2. The model introduces a “safe” action ($a = 1$) with payoff S and a “risky” action ($a = 2$) with unknown payoff R or r . What does Assumption A1 (Safe Default) guarantee, and why does it matter for the setup?
 3. The parameter v represents the agent’s prior probability that the risky payoff is R (the good outcome). Under A1, the agent chooses the safe action without any information. Why?
 4. What is $\hat{\lambda}$, and how does Gans interpret it as a measure of the “quality of judgment”?
 5. In Section 5.1.3, Gans describes two judgment strategies: $J1$ (judge one state only) and $J2$ (judge both states contingently). In plain language, what is the difference between them?
 6. Why is it never optimal for the agent to apply judgment to a second state unconditionally (i.e., regardless of what was learned about the first state)?
 7. What does Assumption A2 (Judgment Insufficient) say, and why does Gans impose it?
 8. Proposition 5.1 gives conditions under which prediction and judgment are complements. State the key condition in your own words.
 9. When $\hat{\lambda}$ is relatively high (the agent has high-quality judgment), prediction and judgment can become **substitutes** rather than complements. In one or two sentences, explain the intuition for why this happens.
 10. In one sentence, what is the main takeaway of Sections 5.1–5.2 for understanding the demand for AI prediction?